



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°3_ – RÉSEAUX 2

COURS: RÉSEAUX 2

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NIVEAU: L3

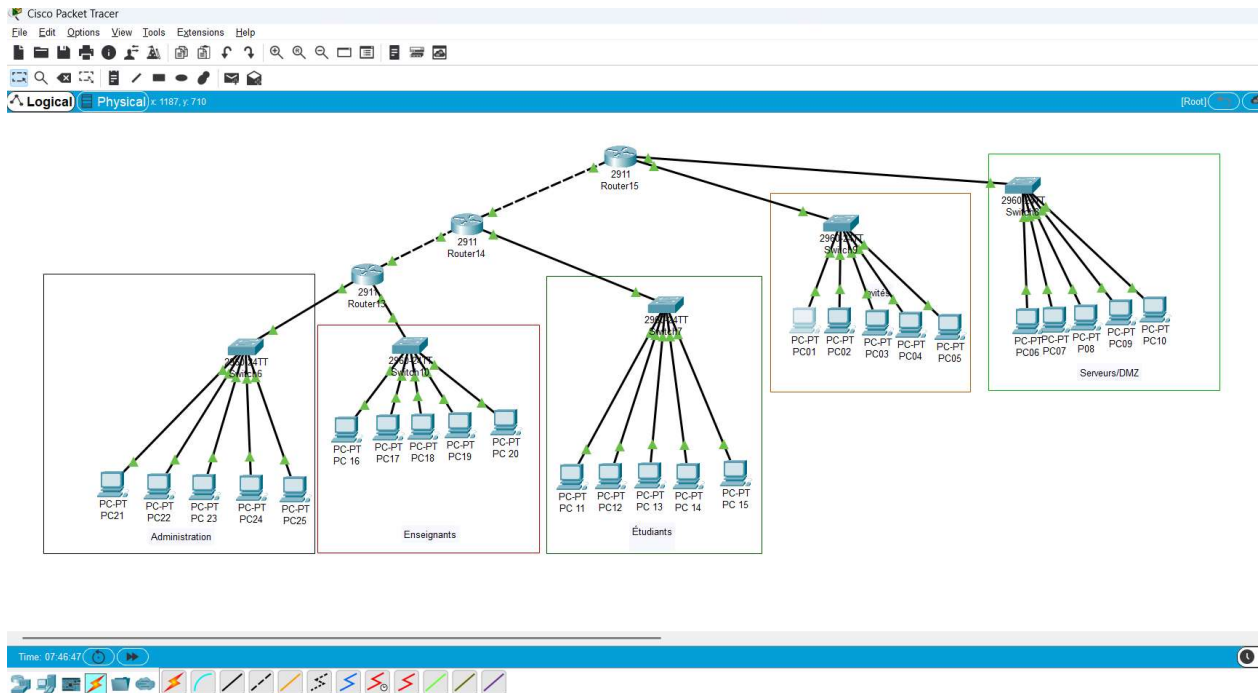
DATE: 04/04/2026

L'objectif de ce TD est de:

1. Apprendre à créer et gérer plusieurs VLANs pour isoler les différents départements (Administration, Commercial, Technique, Invités, DMZ).
2. Comprendre le routage inter-VLAN à l'aide de routeurs et de sous-interfaces.
3. Identifier les avantages de la segmentation pour la sécurité.
4. Configurer des mots de passe forts pour l'accès console et VTY.
5. Activer SSH sur les routeurs pour un accès sécurisé à distance.
6. Appliquer la port-security sur les switchs pour limiter les adresses MAC autorisées sur chaque port.
7. Créer des ACL étendues pour contrôler l'accès entre VLANs (ex. : VLAN invités ne peuvent pas accéder à la DMZ ou aux ressources sensibles).
8. Filtrer le trafic réseau de façon sélective (protocole, IP source/destination).
9. Simulation de scénarios d'attaque et test

1. Créer la topologie réseau

- Pour cette topologie j'ai utilisé des routeurs donnant l'accès à trois interfaces.



2. Configurer les VLANS

Admin

```

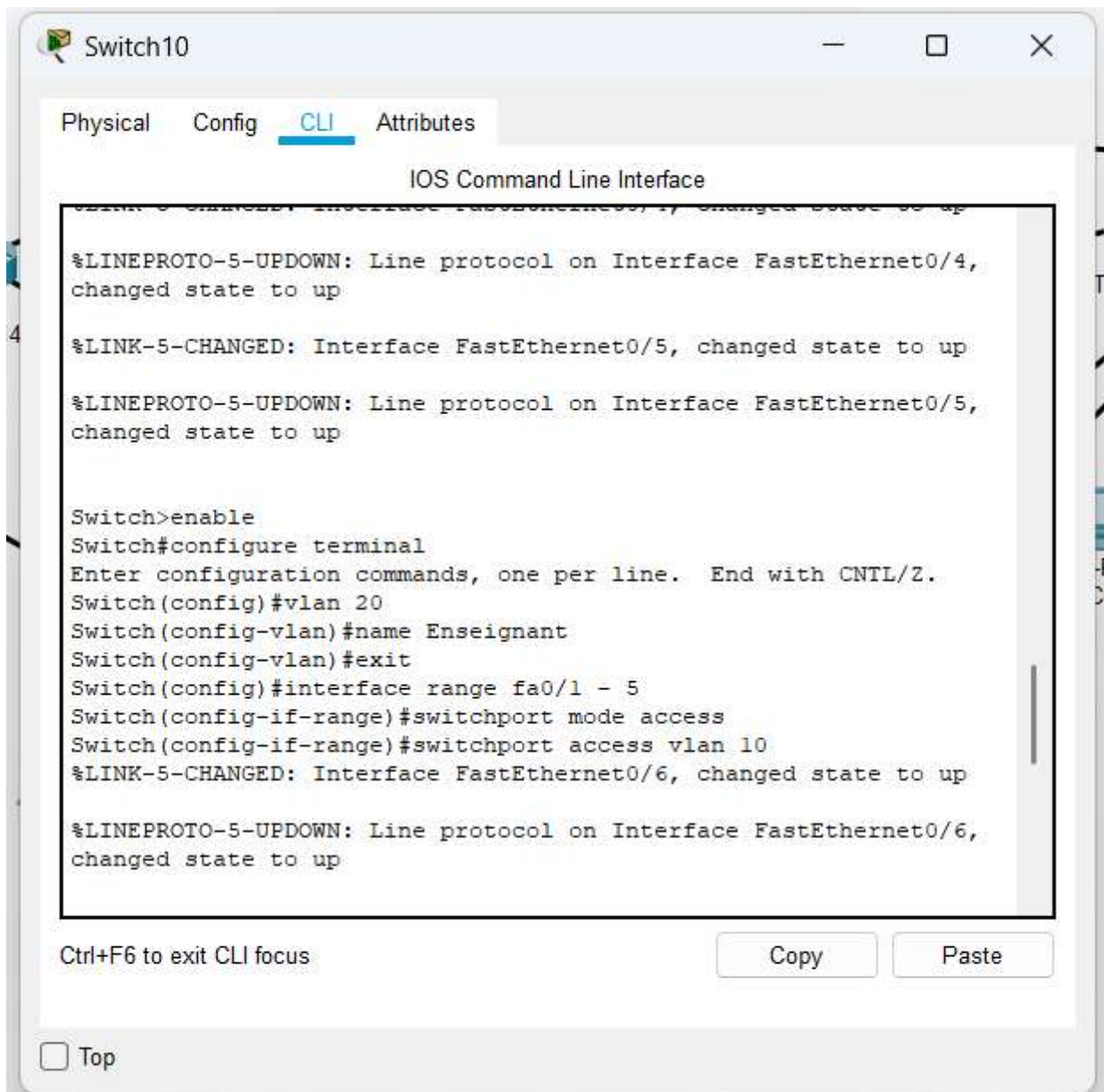
Switch6
Physical Config CLI Attributes
IOS Command Line Interface
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6,
changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Administration
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

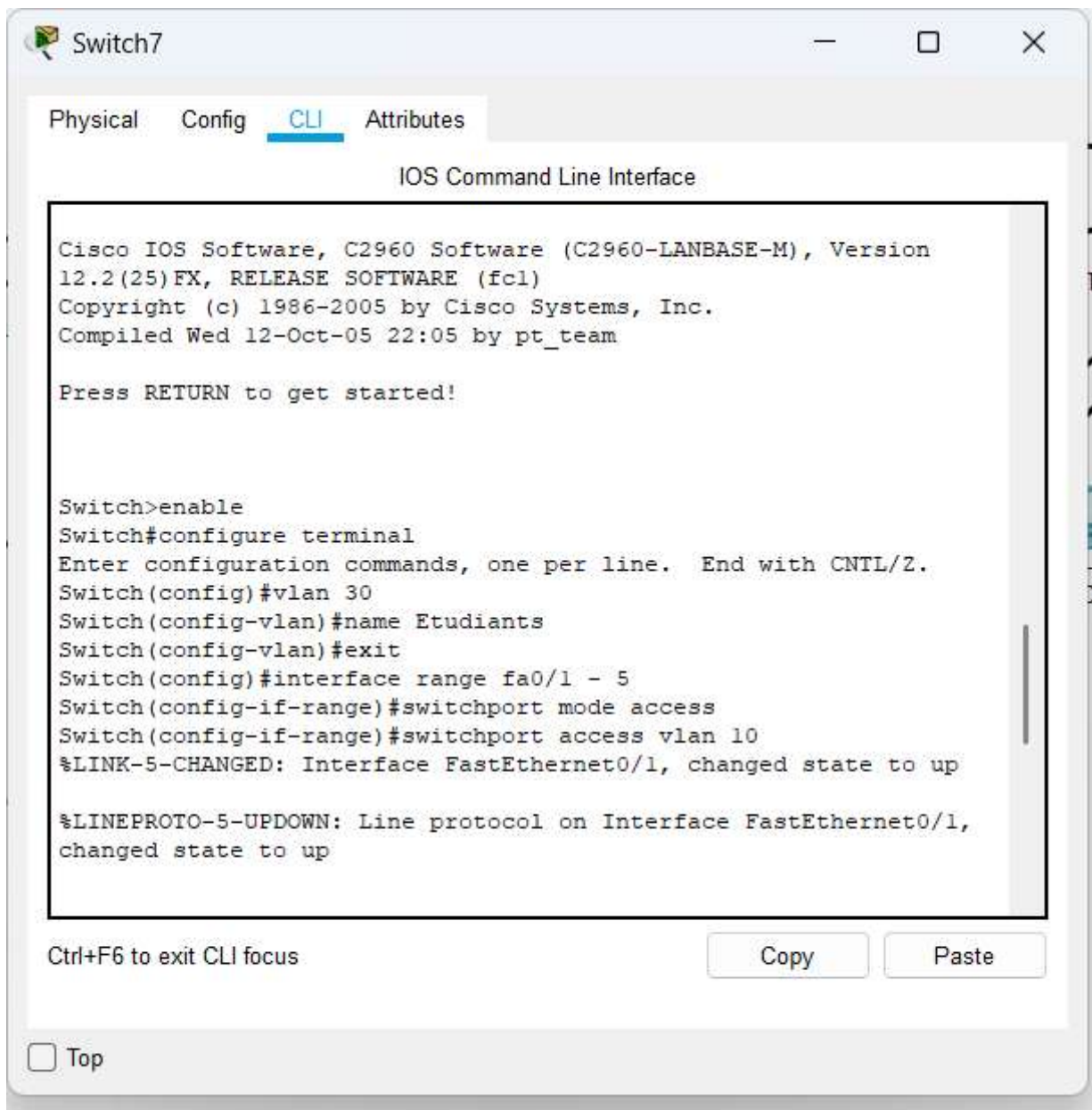
Ctrl+F6 to exit CLI focus
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Top

```

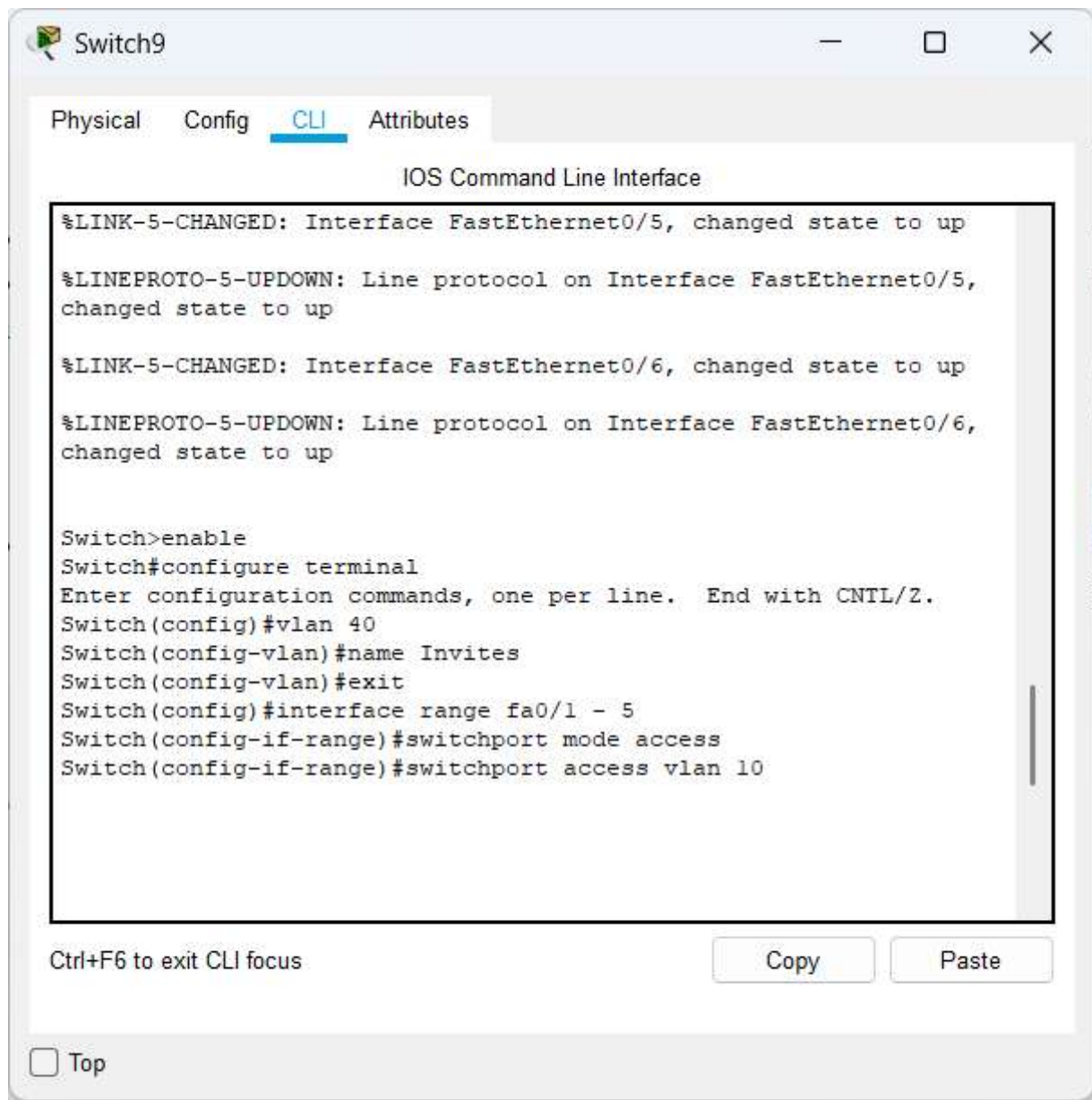
Enseignants



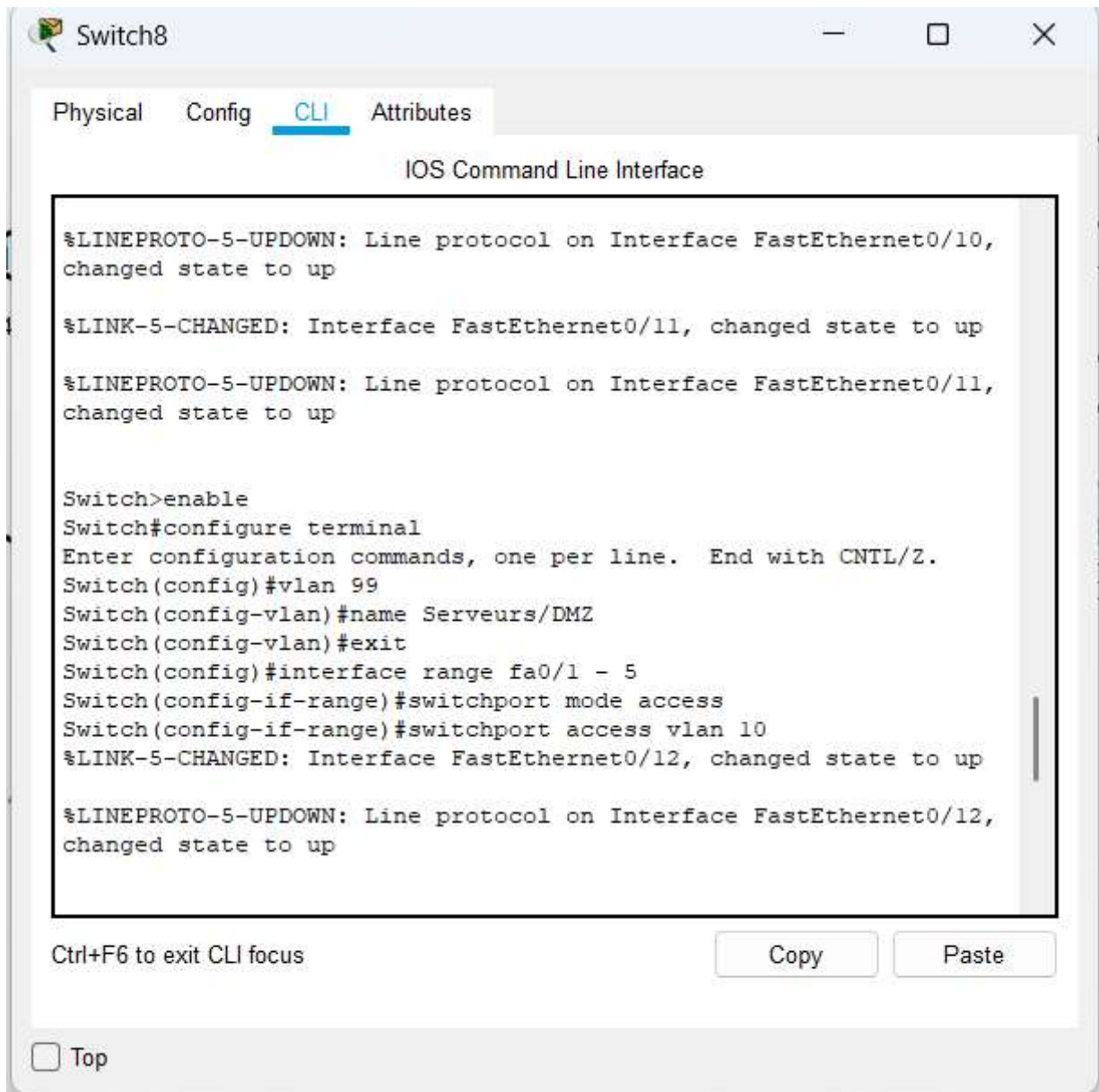
Etudiants



Invités



Serveurs DMZ



3. Configurer le routage inter-VLAN



Router13



Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to
up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
up
```

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Top



Router14

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/2, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.40.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.50.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.60.1 255.255.255.0
Router(config-if)#no shutdown
```

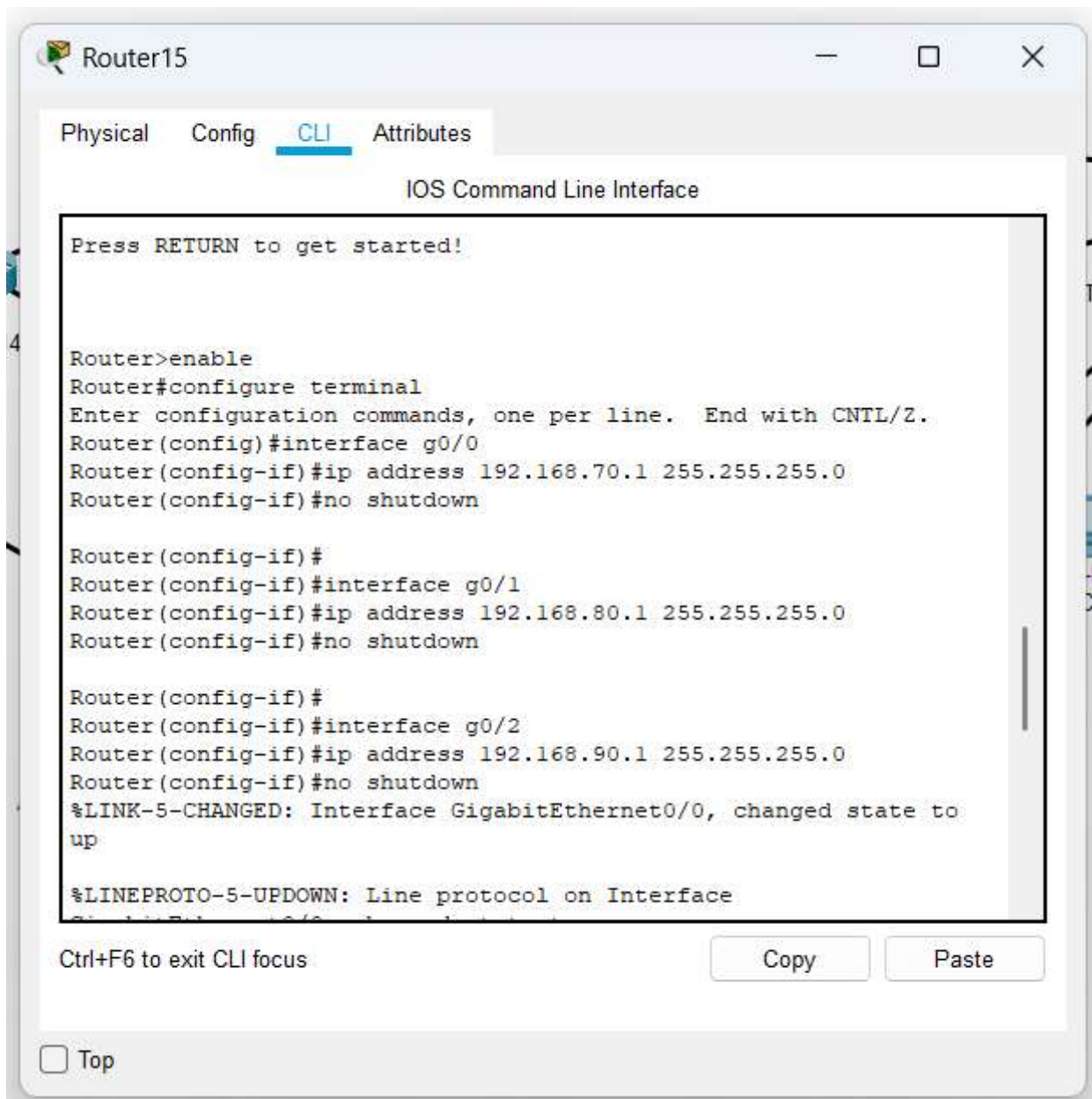
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4. Mettre en place la sécurité réseau

A- ACL (Access Control List)

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255
192.168.99.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#
```

Ctrl+F6 to exit CLI focus

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B- Port Security

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#
```

Ctrl+F6 to exit CLI focus

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Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started.

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#
```

Ctrl+F6 to exit CLI focus

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☐ Top

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#
```

Ctrl+F6 to exit CLI focus

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☐ Top

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#
```

Ctrl+F6 to exit CLI focus

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☐ Top

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#
```

Ctrl+F6 to exit CLI focus

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C- Sécurisation des routeurs

```
Router15#enable
Router15#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router15(config)#hostname Router13
Router13(config)#ip domain-name campus.local
Router13(config)#crypto key generate rsa
% You already have RSA keys defined named Router15.campus.local .
% Do you really want to replace them? [yes/no]: 1024
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: username admin secret Cisco@123
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: line console 0
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: password Cisco@123
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: login
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: line vty 0 4
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: transport input ssh
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: login localy
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: Router13.campus.local
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Router13(config)#
```

```
Router15#enable
Router15#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router15(config)#hostname Router14
Router14(config)#ip domain-name campus.local
Router14(config)#crypto key generate rsa
% You already have RSA keys defined named Router15.campus.local .
% Do you really want to replace them? [yes/no]: 1024
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: username admin secret Cisco@123
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: line console 0
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: password Cisco@123
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: login
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: line vty 0 4
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: transport input ssh
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: login locally
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: Router14.campus.local
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Router14(config)#
```

Ctrl+F6 to exit CLI focus

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router15
Router15(config)#ip domain-name campus.local
Router15(config)#crypto key generate rsa
The name for the keys will be: Router15.campus.local
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

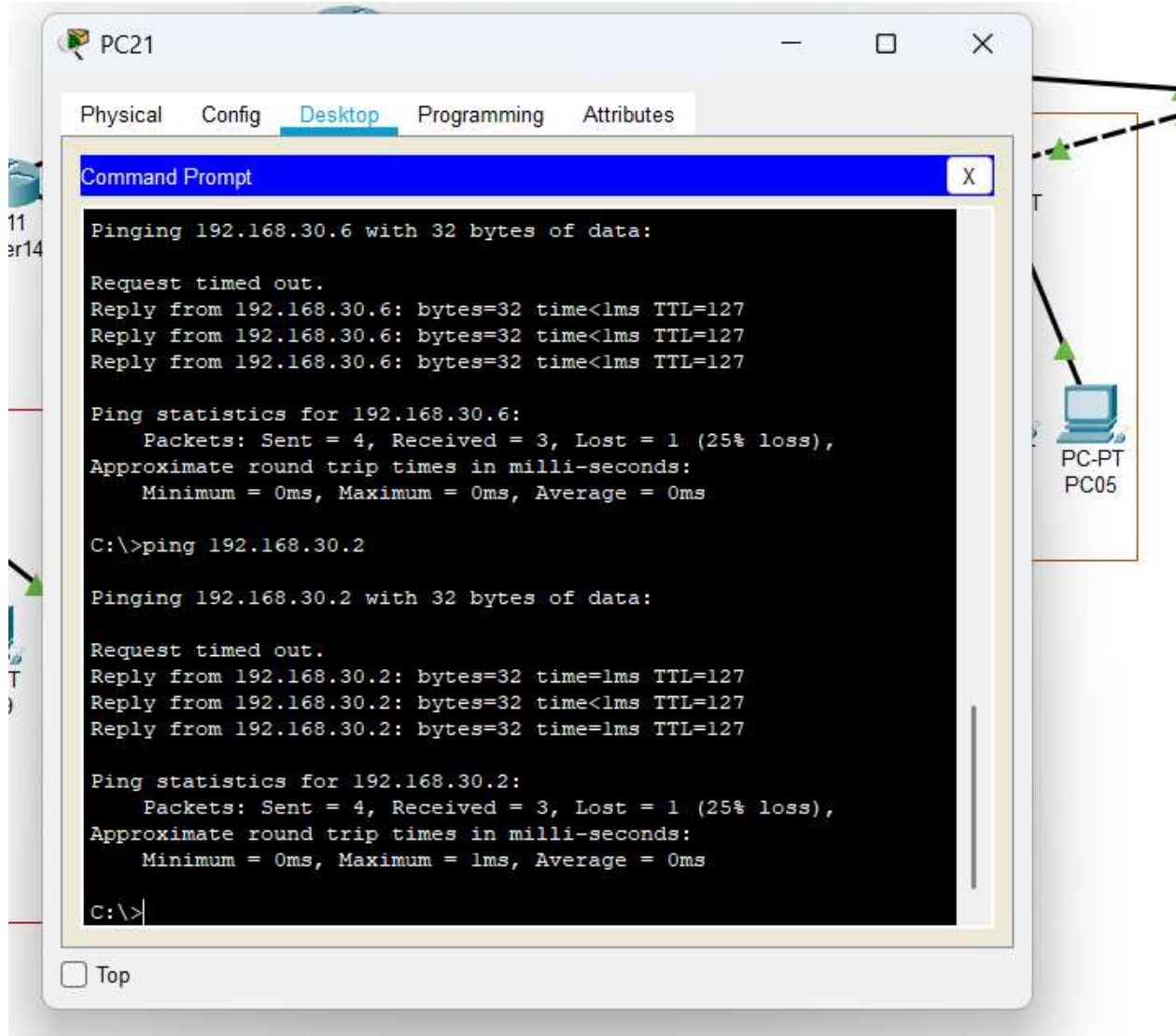
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Router15(config)#username admin secret Cisco@123
*Mar 1 1:44:15.94: %SSH-5-ENABLED: SSH 1.99 has been enabled
Router15(config)#line console 0
Router15(config-line)#password Cisco@123
Router15(config-line)#login
Router15(config-line)#line vty 0 4
Router15(config-line)#transport input ssh
Router15(config-line)#login local
Router15(config-line)#
```

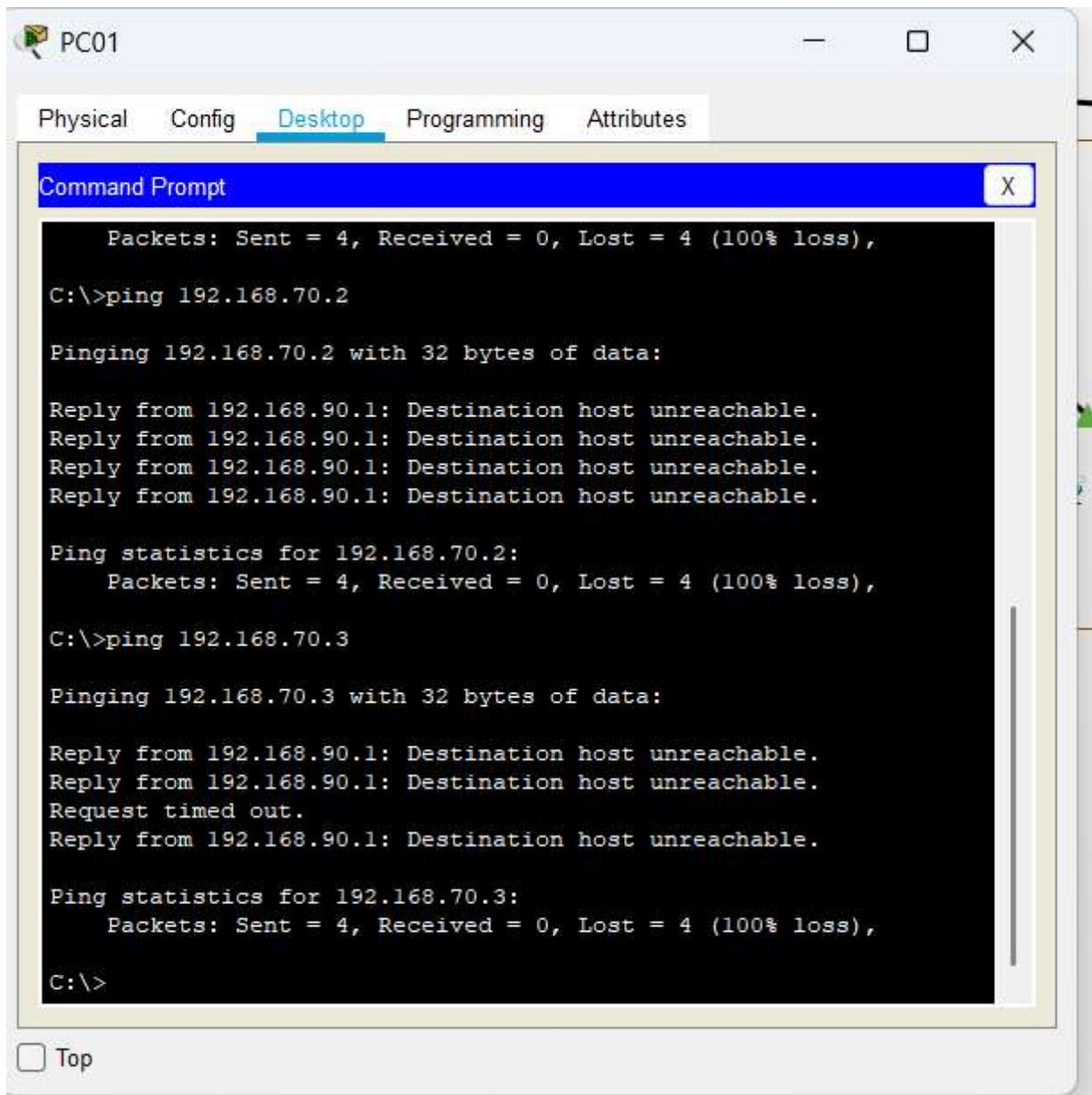
Ctrl+F6 to exit CLI focus

Scénarios de test pour le TD

1. Ping entre VLAN autorisés : admin et enseignants fonctionne.



2. Ping VLAN invités et serveur DMZ échoue.



- Bloqué par l'ACL

3. SSH routeurs : tester que Telnet est désactivé et SSH fonctionne.

```

C:\>ssh -l admin 192.168.30.1

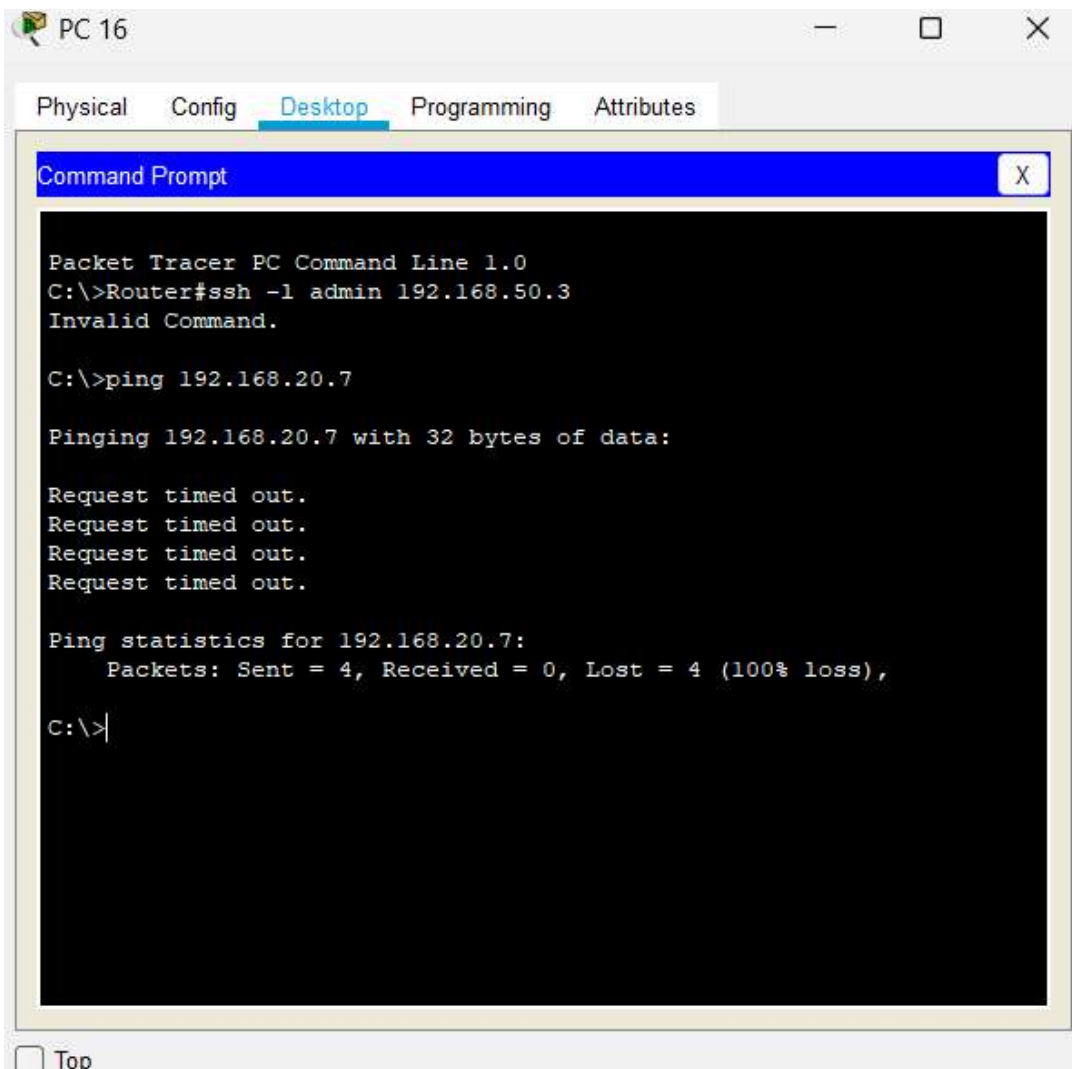
Password:

Routerl3>exit

[Connection to 192.168.30.1 closed by foreign host]
C:\>telnet 192.168.30.2
Trying 192.168.30.2 ...
% Connection refused by remote host

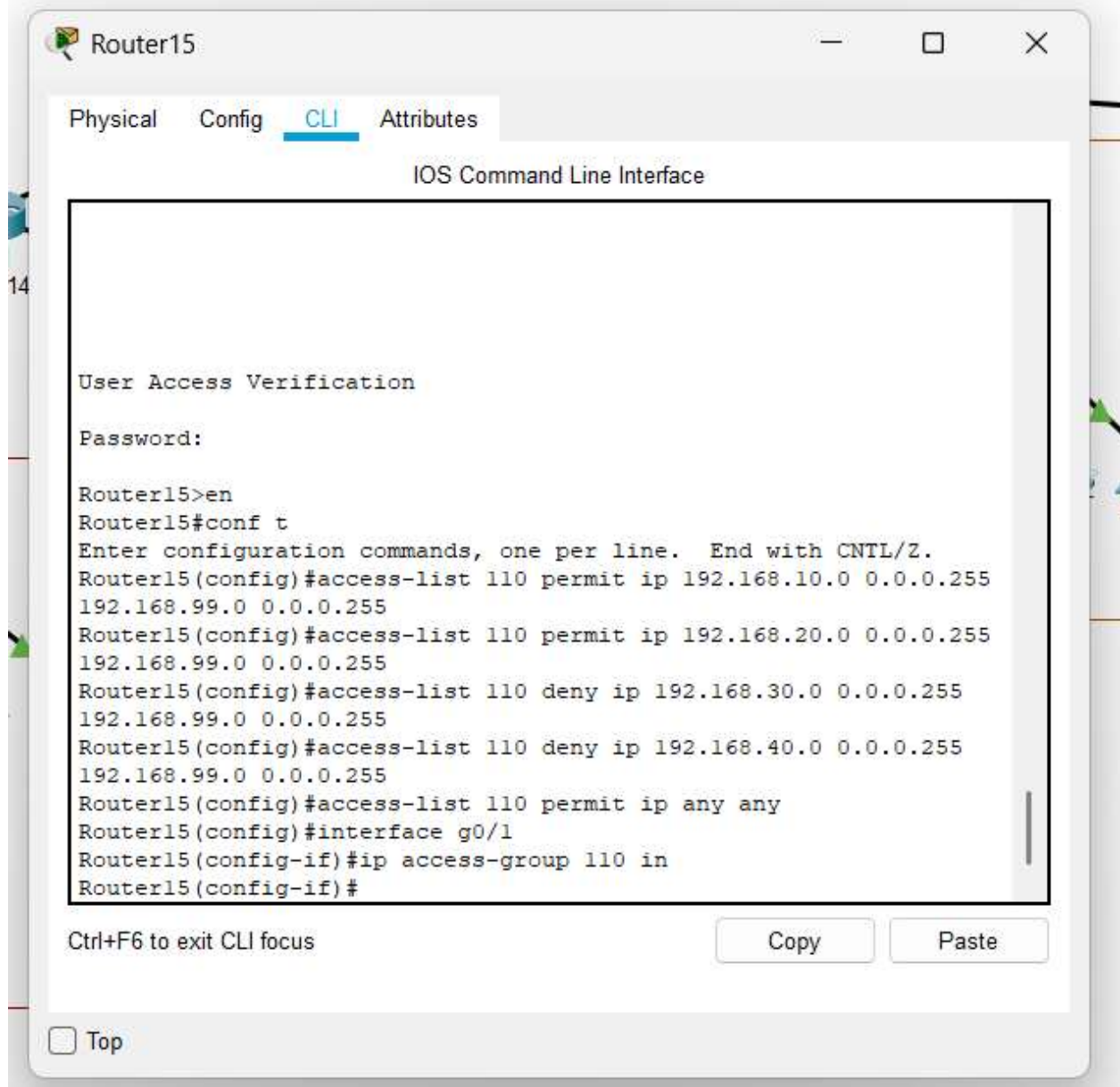
```

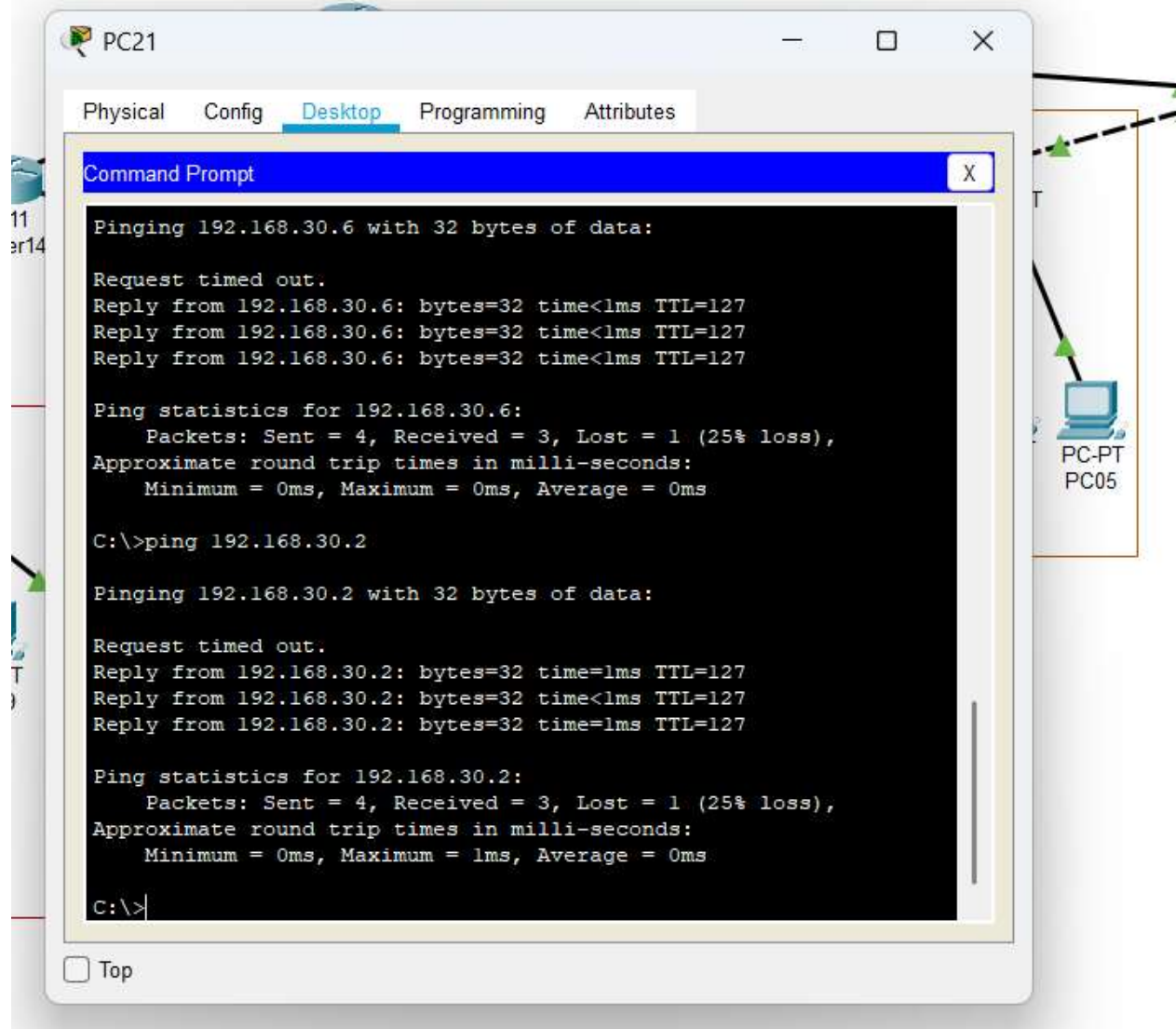
4. Port Security : connecter un 3e PC sur un port limité doit être restreint.

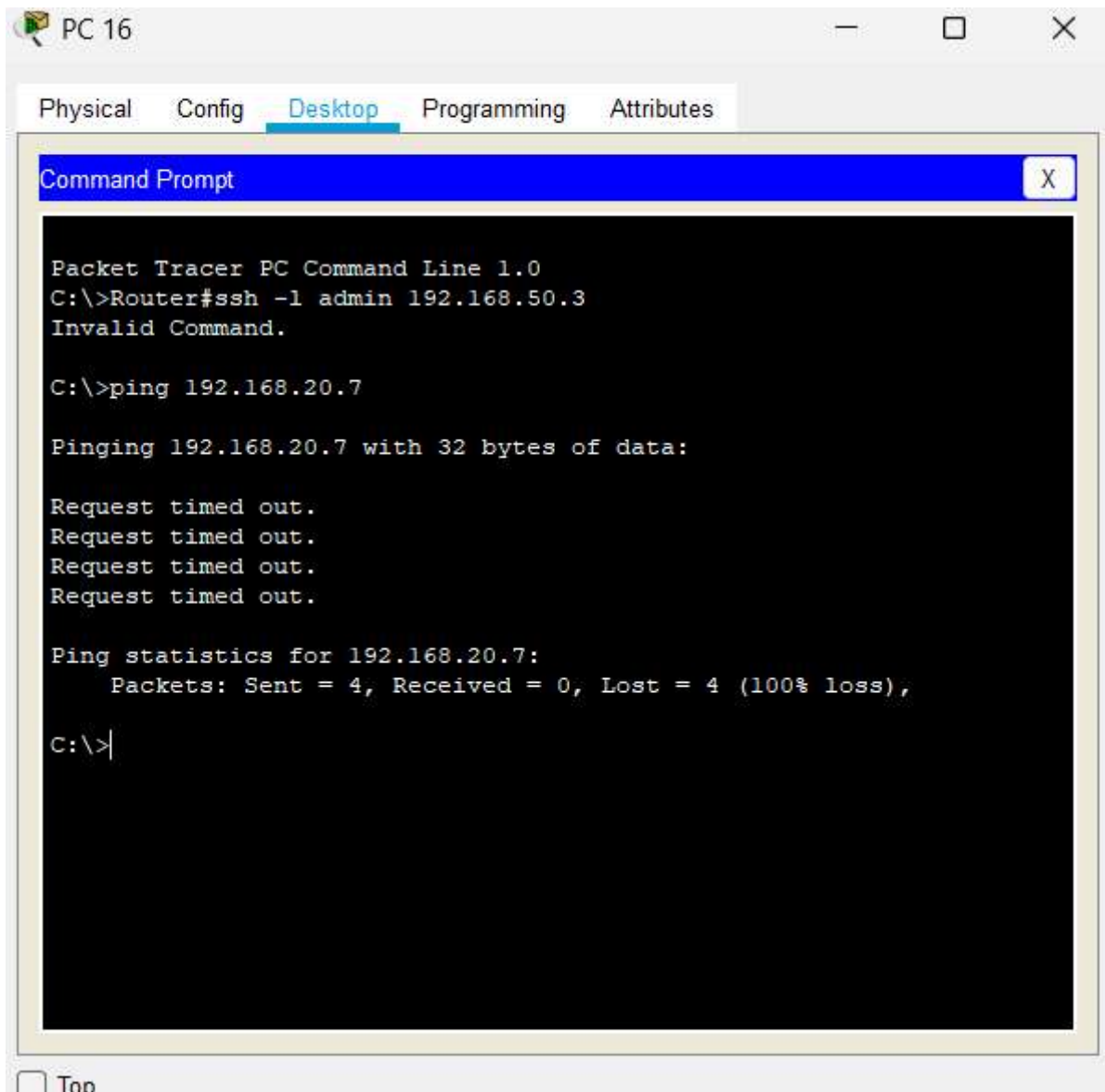


- J'ai ajouté un 6ème pc sur mon port dont le nombre de PC est limité le ping est échoué car le port security bloque le PC supplémentaire.

5. Accès aux serveurs : seuls admin et enseignants autorisés.







Cocclusion.

Ce TD m'a permis de comprendre comment construire un réseau segmenté en VLANs, sécurisé par ACL et Port Security, administré uniquement en SSH avec mots de passe forts, puis tester avec des scénarios pour prouver que la sécurité fonctionne. donc la tache a été réussie.